What does “on average” mean? And how much of an improvement can we get?

There are two ways to answer these questions: from a theoretical point of view, we can analyze the expected height of a randomized treap, and mathematically prove that, on average, the height will be logarithmic.

But also, from a practical angle, we can just run a simulation to verify that what we expect is true, and compare the height of a BST versus a randomized treap with the same elements.

3.5.1 Theory: Expected Height

To analyze the expected height of a random data structure, we need to introduce some concept about statistics.

First and foremost, we will need to use the concept of expected value for a *random variable* `V`: we can informally define it as the mean value (**not** the most likely value) that variable will assume over a large set of occurrences.

More formally, if `V` that can assume values in a finite, countable set `{v1, v2, … vM}`, each with probability `{p1, p2, …, pM}`, then we define the expected value of `V` as:

$$E[V] = \sum_{i=1}^{M} v_i \cdot p_i$$

For our randomized treap, we will define a random variable `DK` for the depth of a given node `NK`, where the index `k ∈ {0,…n-1}` denotes the index of the node’s key in the sorted set: `NK` has the `k`-th smallest key in the tree.